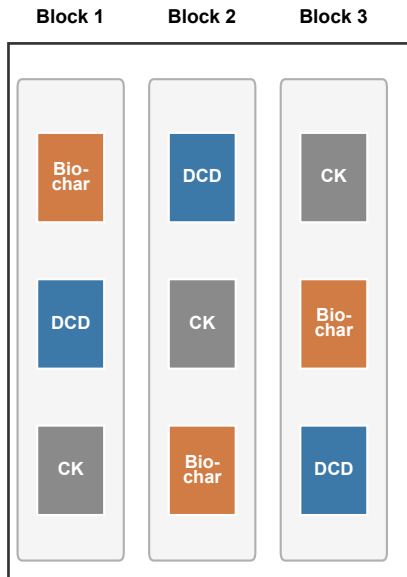


a

One existing configuration-site unit (CSU)

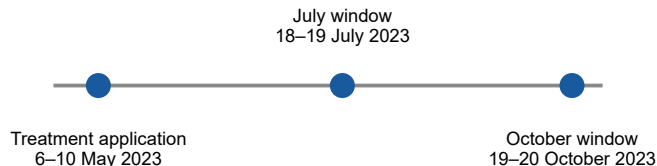
80 m × 5 m strip; 11 CSUs in total



One CK, one DCD and one biochar plot per block;
order randomized independently within blocks

b

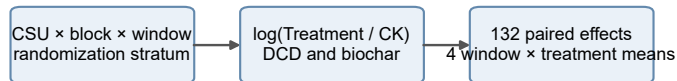
Field timeline and repeated measurements



99 permanent collars × 2 windows = 198 original flux observations

c

Paired effect construction and inference boundary



Primary inference CSU-clustered covariance (11 clusters; t, df = 10)

Sensitivity Randomized-block clustering (33 clusters; t, df = 32)

Inference boundary CSUs are existing contexts, not randomized plant treatments;
CSU-specific estimates are descriptive and are not ranked by significance.